Unit 5 common confusions reference

public-student | All competition levels

Confusion	Correction and reason
Kelvin and Black both discovered heat.	Kelvin developed absolute-temperature ideas; Black clarified specific and latent heat. Neither broad phrase captures their separate contributions.
Same temperature means the same thermal energy.	Mass, material and phase matter. A thermometer alone does not provide an energy total.
A melting plateau means no heat enters.	Energy can drive a phase change with little temperature change.
Joule’s paddles supply heat through the shaft.	For the water system the shaft supplies work. Warming describes the result, not the transfer mode.
Carnot wrote today’s complete energy theory in 1824.	His published framework used caloric assumptions; later thermodynamics revised the energy interpretation.
Any engine below 100% is feasible.	It must also respect the reservoir-dependent Carnot bound and other physical constraints.
Clausius and Boltzmann both first named entropy.	Clausius named entropy in 1865; Boltzmann developed its statistical interpretation, notably in 1877.
Entropy is energy that was destroyed.	Entropy is a state property in J/K. Energy is conserved even when useful work potential decreases.
Every molecule at one temperature has the same speed.	Temperature describes collective behavior; equilibrium gases have distributions.
Maxwell–Boltzmann proves a shared paper.	The name recognizes connected contributions and is not evidence of coauthorship.
The demon is a proven perpetual-motion machine.	It is a thought experiment. Include the physical controller and repeated restoration in the full accounting.
Planck invented the word entropy.	His key contribution here concerns radiation and energy quanta; Clausius introduced entropy.
1900 and 1918 are interchangeable Planck dates.	1900 labels the breakthrough; 1918 labels the Nobel Prize.
Blackbody means no emission.	It is an ideal absorber and thermal emitter; sufficiently hot examples glow.
A diagram proves historical collaboration.	A conceptual map can show related ideas without documenting personal interaction.